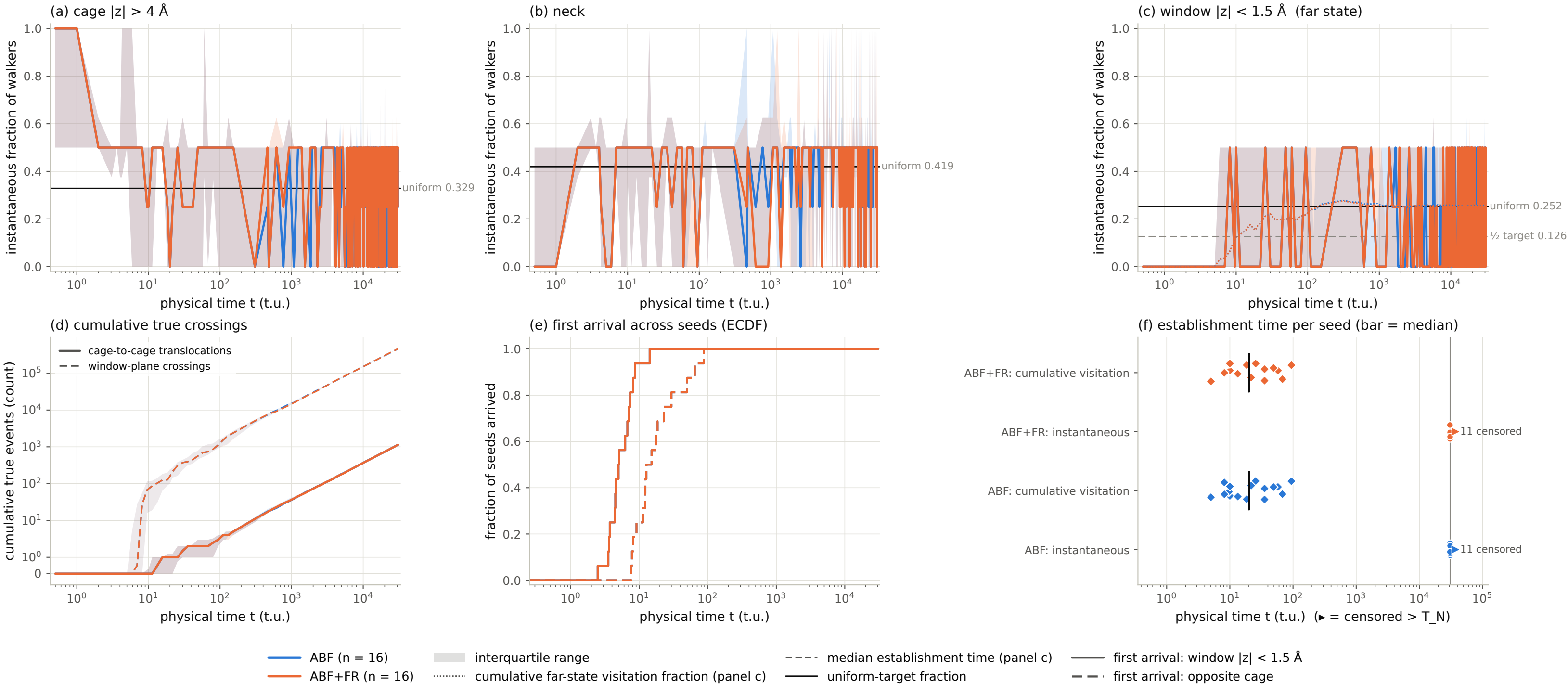


**Ethane/LTA 300 K · N = 2 · T\_N = 30720 t.u.**

n\_steps = 153,600,000 · h = 0.0002 · B = N n\_steps = 3.072e+08 walker-steps · complete seeds: ABF 16/16, ABF+FR 16/16

Establishment and crossings vs physical time



Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC\_PLAN section-4 quantities, unsmoothed.

Establishment (frozen): first save after which the instantaneous far-state fraction (window  $|z| < 1.5 \text{ \AA}$ ) stays  $\geq \frac{1}{2}$  of its uniform target 0.252; the cumulative-visitation version uses C\_all (far-state bins) against  $\frac{1}{2}$  of their bin share. True events never count a copy (clones inherit labels).

$N \leq 2$ : instantaneous establishment is ill-defined (each walker is half the population or more); read the cumulative-visitation version.

Null calibration at  $N = 2$ : an EXACTLY uniform population (far count  $\sim \text{Bin}(N, 0.252)$  per save) fails the instantaneous criterion at a save with  $p = 0.56$  and passes every save of the second half with  $P = 0.000 \rightarrow$  UNRELIABLE here: the instantaneous establishment time is censored / late by chance even when established, and FR birth-death can push the count below binomial scatter (an FR advantage in it is not evidence of faster convergence); read the cumulative version and the first arrival.

eqb\_plot\_config/2 · config lta\_300K\_production.json · 2026-10-10 09:47 UTC